# TITLE OF THE WORK :DRONE BASED TRANSMISSION LINE AND SUBSTATION INSPECTION USING AI


## ABSTRACT

The reliability of power transmission and distribution systems depends largely on the continuous monitoring of **transmission lines and substations**. Traditional inspection methods, such as manual patrols and helicopter surveillance, are often **time-consuming, costly, and hazardous**, while offering limited real-time monitoring capabilities. To address these challenges, this project proposes a **drone-based fault inspection and monitoring system** that integrates **Raspberry Pi, ESP32, IoT-enabled sensors, and image processing techniques**.

In the proposed system, a drone is equipped with a **Raspberry Pi and camera module** to capture real-time images and videos of transmission infrastructure. **Image processing algorithms** are employed to detect visible faults such as insulator cracks, conductor sagging, and flashover marks. Simultaneously, **sensors** like current, voltage, temperature, vibration, and partial discharge are connected through Arduino provide electrical and environmental condition monitoring. The **ESP32 module** enables wireless IoT communication, transmitting the processed result including fault type, sensor readings, and GPS location to a **cloud-based dashboard**. Operators can then access live monitoring data, receive **instant alerts**, and pinpoint fault locations for quick maintenance response.



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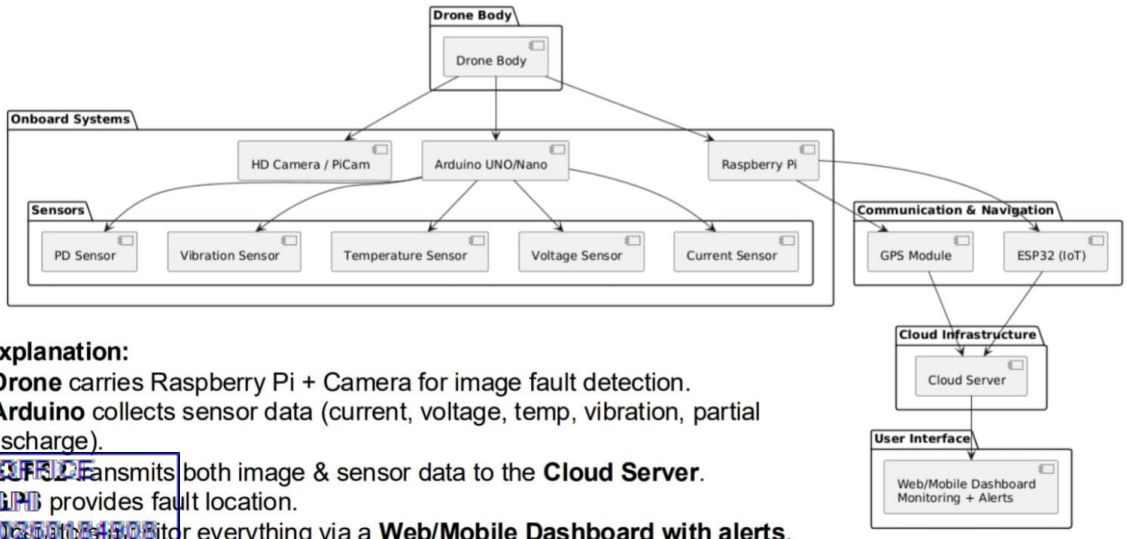
## OBJECTIVES

- **drone-based aerial inspection system** for capturing live video/images of transmission lines and substations.
- **Raspberry Pi with image processing (OpenCV/TensorFlow Lite)** for detecting visible faults such as insulator cracks, conductor sag, flashover marks, and vegetation intrusion.
- Implement **ESP32-based IoT communication** for wireless transmission of fault data (images, sensor values, GPS location) to the cloud.
- develop a **cloud dashboard** for live monitoring, fault visualization, and alert generation.

## SCOPE

- Real-time **visual fault detection** using a drone-mounted camera and Raspberry Pi.
- **Electrical condition monitoring** using IoT-based sensors (current, voltage, temperature, vibration, partial discharge).
- **Wireless data transmission** using ESP32 to a cloud platform.
- **Dashboard interface** for operators to monitor faults, view images, analyse sensor readings, and receive alerts.
- **based fault localization** for quick maintenance response.

## WORKING OF PROJECT




**Explanation:**

•**Drone** carries Raspberry Pi + Camera for image fault detection.

•**Arduino** collects sensor data (current, voltage, temp, vibration, partial discharge).

•ESP32 transmits both image & sensor data to the **Cloud Server**.

•Cloud AI provides fault location.

•Operator monitor everything via a **Web/Mobile Dashboard with alerts**.

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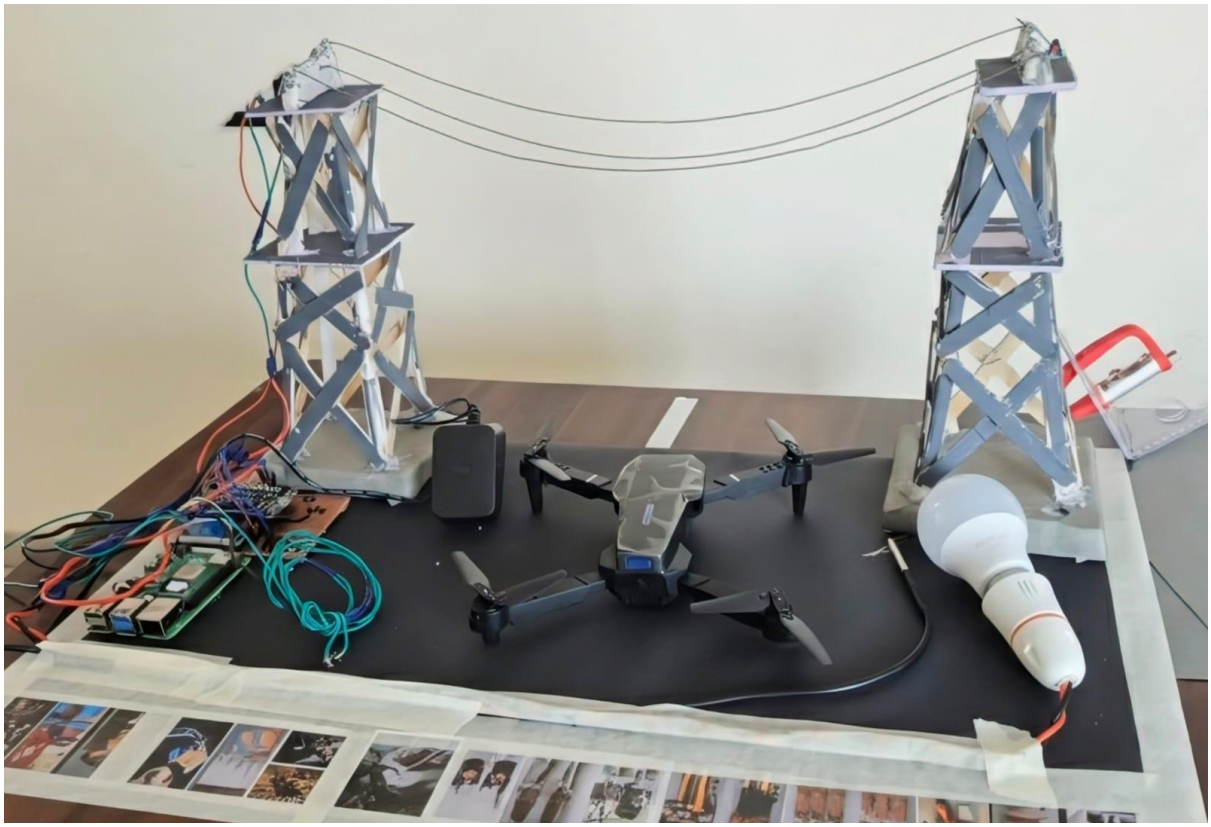
## DESIGN OF PROJECT